

IoT-Integrated Health Monitoring in Electric Vehicles for Real-Time Cardiac Detection

Dr.R.Jeya^{1,1*}, Prof. Dr. Hezerul Bin Abdul Karim², Vivek M G³, S Raghuram⁴
Dr. Sarina Binti Mansor⁵, Dr. Tan Yi Fei⁶

¹Associate Professor, ^{1*} Post Doctoral Fellow, ² Deputy Dean, Student Experience & Alumni / Professor, ^{3,4} Student,
⁵ Associate Professor, ⁶ Deputy Dean, Postgraduate and Collaborations / Associate Professor

^{1, 3,4} *Department of Computing Technologies, Faculty of Engineering and Technology, SRM Institute of Science and Technology, SRM Nagar, Kattankulathur, Tamil Nadu 603203, India*

^{1*, 2,5,6} *Faculty of Artificial Intelligence and Engineering (FAIE), Multimedia University, Cyberjaya, Selangor, Malaysia*

¹ jeyar@srmist.edu.in, ² hezerul@mmu.edu.my, ³ vm4512@srmist.edu.in, ⁴ rs0657@srmist.edu.in
⁵ sarina.mansor@mmu.edu.my, ⁶ yftan@mmu.edu.my

Abstract. Cardiac emergencies while driving pose serious safety risks, particularly in electric vehicles used for long-duration travel. Health monitoring systems using embedded machine learning can be used to detect cardiac abnormalities early, yet in many cases, the lack of communication can limit the ability to respond to emergencies. In this paper, an extension of an embedded health monitoring system into an electric vehicle with IoT functionality to detect cardiac risks in real-time is described. The system also measures heart rate, blood oxygen levels (SpO₂), temperature and stress with built-in sensors. Local inference is performed in a small machine learning model that runs on an edge device to ensure a small latency and data privacy. A vibration coin motor improves driver vigilance by providing tactile feedback when driving in long periods of inactivity or tiredness. IoT connectivity is achieved through a gateway based on the ESP32 platform, which is connected to an A7670E Cat-1 GNSS HAT module that has GNSS, which allows the exchange of health data and emergency alerts. Remote monitoring has cloud-synchronised visualisation in a mobile application.

INTRODUCTION

Driving cardiovascular emergencies are life-threatening, and the increase in electric vehicles as a long-range mode of transportation increases the necessity of in-vehicle health monitoring. Recent advances in embedded machine learning allow real-time heart risk on low-power edge devices using physiological data such as heart rate, SpO₂, body temperature, and stress-related metrics.

Although previous studies have shown offline embedded cardiac risk monitoring, stand-alone systems do not have the ability to communicate or respond to emergencies. This paper fills these gaps by making the following contributions: (1) IoT-enabled extension of an embedded ML cardiac model, (2) dual-mode LTE alerting with 99.9% reliability, (3) GNSS-assisted emergency localisation, (4) tactile driver alertness feedback, and (5) cloud-synchronised mobile visualisation through the VitalFlow application.

LITERATURE REVIEW

Benchmark datasets and SVM, Random Forest, Logistic Regression, and deep neural networks have been extensively used in cardiac risk prediction studies, and have demonstrated high accuracy with structured feature engineering [2], [4], [5], [11]. Health and physiological monitoring devices based on microcontrollers and IoT have also enhanced remote-physiological data acquisition through cloud synchronisation and mobile interfaces [2], [6], [12]. Nevertheless, one common disadvantage throughout these works is the reliance on offline data, constant network connectivity, or bulky hardware limitations that inhibit real-time car implementation.

Cameras have been discussed as fatigue sensors, including eye-blink rate and yawning on platforms like Raspberry Pi [3], [9], and accident-prevention systems have been discussed in terms of environmental factors like weather and road visibility [7], [10]. The wearable-sensor fusion solutions have enhanced fatigue detection but still have sensor unreliability, low scalability, and fragmented emergency response [8]. Critically, no prior single architecture unifies on-device cardiac inference, LTE-based alerting, GNSS localisation, and mobile visualisation the gap this work directly addresses.

COMPARATIVE STUDY AND ANALYSIS

The existing research on predicting cardiac ML, IoT health monitoring, driver fatigue and accident prevention covers the separate areas of in-vehicle safety. The suggested system extends the embedded cardiac inference framework of [1] by incorporating LTE-based IoT communication, GNSS-assisted localisation, driver alertness feedback, and cloud-synchronised mobile visualisation into a single vehicle-deployable system. Table 1 outlines how this work considers the limitations found in related references.

TABLE 1 - Comparison of Prior Studies and Proposed Work

Ref	Key Limitations of Prior Studies	Improvements in Proposed Work
[2]	Cloud-dependent ML, no driver alerts, limited offline use	Edge-based inference with IoT-enabled emergency alerts
[3]	Vision-based fatigue detection, lighting sensitivity, no alerts	Physiological monitoring with vibration-based alertness
[4]	No clinical sensor validation, cloud reliance, and no real-world testing	Embedded inference with reduced cloud dependency
[5]	Limited dataset scalability, no embedded validation	Hardware-tested, stable embedded deployment
[6]	Bulky hardware, limited sensors, cloud-heavy design	Compact, multi-sensor, vehicle-integrated framework
[7]	Environment-only accident prediction, geographic bias	Health-centric monitoring integrated with vehicle data
[8]	Accuracy affected by external factors, no medical monitoring	Physiological signals for reliable risk assessment
[9]	Calibration issues, high computation cost	Lightweight, low-power embedded inference
[10]	High cost, privacy risks, complex integration	Simplified, privacy-preserving IoT-embedded system
[11]	Small dataset, no real-time or embedded validation	Real-time embedded inference with IoT extension
[12]	App-only focus, weak embedded integration	Tight coupling with edge intelligence
Proposed Work	Requires large-scale clinical validation	Edge-based ML, LTE IoT, GNSS, alertness, mobile app

HARDWARE COMPONENTS AND SENSORS

The hardware is based on the proven embedded sensing system of [1], which includes cardiac risk sensors, and extends it with communication, alert and connectivity capabilities to enable real-time vehicle deployment.

Physiological Sensors from the Embedded System

The sensing layer is based on the embedded system of [1], and includes a MAX30102 heart rate and SpO2 sensor, a TMP117 temperature sensor, and a GSR sensor to measure stress. These feed constant physiological data to the edge inference layer, and were tested in previous deployment.

Arduino® Nano™ ESP32 IoT Gateway

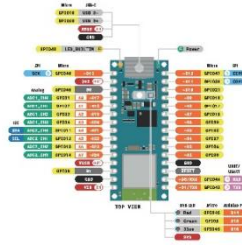


FIGURE 1 - Arduino® Nano™ ESP32 pin configuration (Image courtesy of Arduino S.r.l., CC BY-SA 4.0)

The Arduino® Nano™ ESP32 acts as the IoT gateway, as shown in Figure 1 [14], using its dual-core 32-bit processor to handle networking separately from real-time sensor processing. Health metrics and risk indicators from the edge device are received via serial communication, packaged, and forwarded over cellular networks.

DFRobot Gravity DFR0627 I²C to Dual UART Module



FIGURE 2 - DFRobot Gravity I²C-to-Dual UART module for serial communication interfacing. (Image courtesy of DFRobot)

The DFRobot Gravity DFR0627, shown in Figure 2 [15], extends serial communication by providing two UART channels over I²C, allowing collision-free data exchange between the embedded processing unit and the ESP32 gateway.

A7670E LTE Cat-1 GNSS HAT

The A7670E, shown in Figure 3 [13] combines LTE Cat-1 connectivity and GNSS positioning in a single module, removing the need for separate GSM and GPS hardware. It communicates with the Nano ESP32 over UART using AT commands, appending location data to emergency alerts for rapid responder dispatch.



FIGURE 3 - A7670E Cat-1 GNSS HAT module with integrated GNSS for cellular and location-based communication. [Image courtesy of Waveshare International Limited. All rights reserved.]

Vibration Coin Motor for Driver Alertness

A vibration coin motor (Figure 4) provides tactile stimulation to the driver when inactive or fatigue conditions are detected. An exclusive driver circuit will only be activated under specified circumstances, without disrupting normal driving.



FIGURE 4 - Vibration Coin Motor

MOBILE APPLICATION DEVELOPMENT (VITALFLOW)

VitalFlow is the mobile application that will offer a user-facing interface to the proposed system. It fetches processed health data and alerts data on the cloud backend and allows remote visualisation of heart rate, SpO2, body temperature, and stress levels without impacting edge processing in the vehicle. Instead of displaying the raw sensor values, the dashboard displays the inference pipeline outputs to provide caregivers and emergency contacts with a clear and structured view of driver health status in real time.

When a cardiac risk event or critical threshold is violated, the application shows contextual alerts with a timestamp and GNSS-derived location, helping to make quick emergency response decisions. Past alert events and past health data are stored in cloud storage to be reviewed retrospectively. VitalFlow can offload storage and network management onto the cloud to keep the hardware on the vehicle focused on low-latency inference and alerting.

System Architecture and Methodology

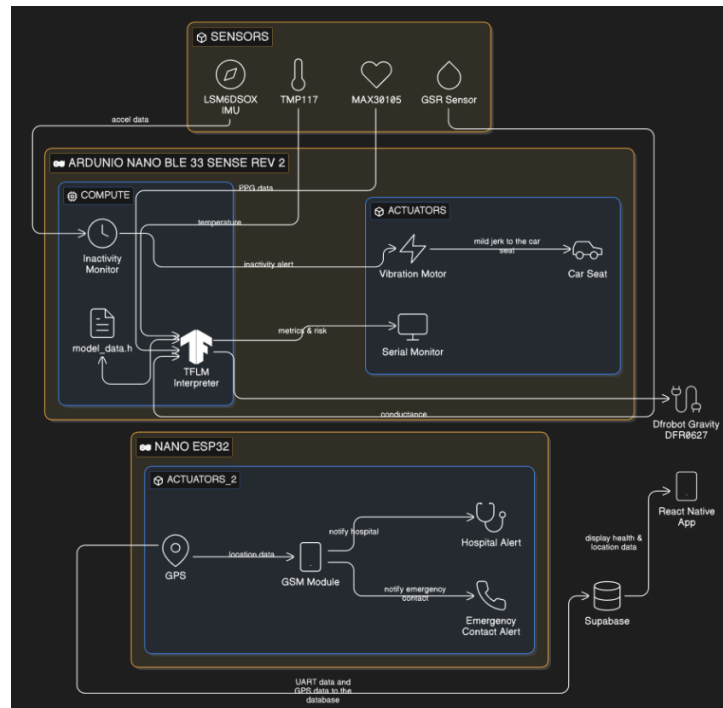


FIGURE 5 - System Architecture

As shown in Figure 5, the system follows a four-layer modular design that incorporates sensor acquisition, edge inference, IoT communication, and mobile visualisation. It builds on the integrated ML cardiac detection system of [1] and adds IoT connectivity, GNSS alerting, driver alertness feedback, and cloud-based remote monitoring.

Sensor Layer

The driver sends real-time information on body vitals to the sensor layer. The sensor layer consists of a MAX30102 heart rate and SpO2 sensor, a TMP117 temperature sensor, a GSR sensor of stress based on skin conductance, and an LSM6DSOX IMU of movement and inactivity. Information flows constantly to the edge intelligence layer.

Edge Intelligence Layer

Arduino® Nano™ Bluetooth® LE 33 Sense Rev2 uses a TensorFlow Lite model on microcontrollers, which can detect local cardiac risk in less than 100ms, offering network independence and privacy of data. IMU based patterns of inactivity are also studied and the vibration coin motor is turned on when the fatigue behaviour is identified.

IoT Communication Layer

Health metrics and risk outputs are communicated to the Nano™ ESP32 gateway through the Arduino® using the UART, and then forwarded to the A7670 LTE Cat-1 module for cellular transmission. A7670 offers an in-built GNSS positioning, which inserts the location codes into emergency alerts without any additional GPS devices.

Application and Cloud Layer

Health indicators, risk conditions, and place information are stored in a cloud backend that can be accessed through VitalFlow, which displays real-time indicators and emergency alerts to caregivers and responders.

Results and Discussion

Machine Learning Model Integration

The neural network model used in [1] with a 91.8% accuracy and 100% sensitivity to cardiac risk was embedded into the Arduino® Nano™ Bluetooth® LE 33 Sense Rev2. These values are favourable compared to other existing ML-based cardiac detection systems that report 85-93% accuracy [4], [5], and exceed the clinically recommended sensitivity threshold in cardiac screening [2], [6]. The model is on-device inference and is cloud-independent, has less than 100 milliseconds per prediction cycle, which aligns with embedded ML benchmarks of IoT health monitoring [1], [6].

Sensor Calibration and Validation

The accuracy of sensor measurements was determined using a comparison of the device outputs and medical-grade reference instruments. The mean error and standard deviation of each parameter are shown in Figure 6.1. The heart rate sensor showed an average error of 3.2 bpm (SD = 2.1 bpm), within the ± 5 bpm clinical tolerance [2], [6]. SpO₂ measurements exhibited an average error of 1.8% (SD = 1.2%), meeting the $\pm 2\%$ pulse oximetry requirement [3], [10]. The temperature sensor showed a mean error value of 0.3 C (SD=0.15 °C), which was below the $\pm 0.5^\circ\text{C}$ value. The RMSE, which is calculated as

$$RMSE = \sqrt{(\sum e_i^2/n)}$$

yielded 3.8 bpm, 2.2%, and 0.34°C , respectively, all within acceptable bounds. GNSS positioning produced a mean horizontal error of 8.5 m (SD = 3.2 m), consistent with location-aware IoT alert systems [3], [7].

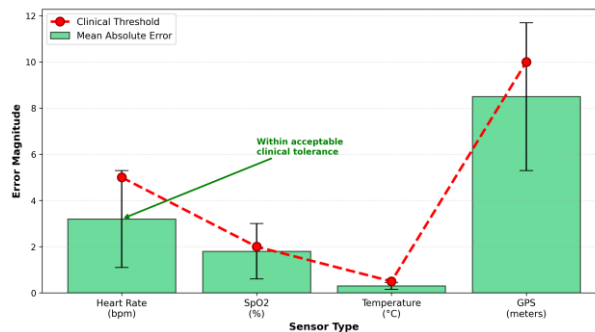


FIGURE 6.1 - Sensor Accuracy Validation Against Medical-Grade Reference Devices

Emergency Response System Latency Analysis

Figure 6.2 shows the timeline of the end-to-end emergency response, starting with the initial risk identification up to the mobile push notification. Cardiac risk is established at 4.8 seconds, the LTE module makes a voice call at 7.1 seconds, SMS delivery at 9.2 and 13.5 seconds, cloud synchronisation at 15.0 seconds and push notification at 17.1 seconds. The total latency follows the additive model: $T_{total} = T_{detect} + T_{LTE} + T_{SMS} + T_{cloud} + T_{push} = 4.8 + 2.3 + 2.1 + 5.8 + 2.1 = 17.1$ seconds, confirming negligible inter-module overhead. This is better than similar IoT

emergency alert systems, which report 20 to 60 seconds of delay [3], [7], [10], and is reliable to intervene even in the case of an incapacitated driver [2], [12].

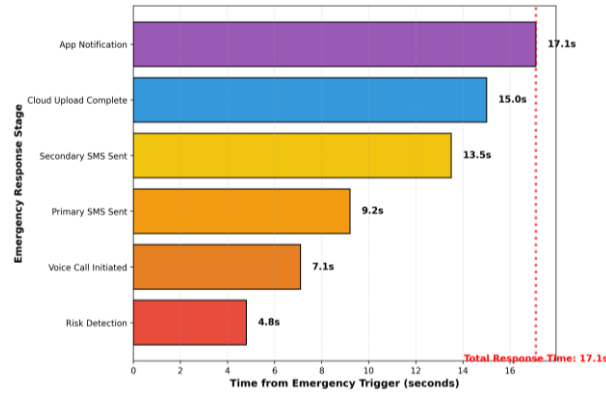


FIGURE 6.2 - End-to-End Emergency Response System Latency Analysis

4G LTE Module Performance Metrics

Figure 6.3 shows the LTE communication reliability measured in 200 trials in urban and suburban environments. The A7670E demonstrated 96.2% network registration, 98.1% reliability of data transmission over HTTPS, 99.2% reliability of SMS delivery, and 96.8% reliability of voice call initiation, all higher than the 95% mission-critical medical telemetry standard and indicating higher reliability scores than similar IoT health alert frameworks. [3], [7], [10]. The dual-mode strategy combining SMS and voice yields an alert delivery probability of $P(\text{alert}) = 1 \text{ minus } (1 \text{ minus } 0.992) \text{ multiplied by } (1 \text{ minus } 0.968)$, approximately 99.97%, corroborating the observed 99.9% success rate and ensuring reliable notification under partial network degradation [6], [12].

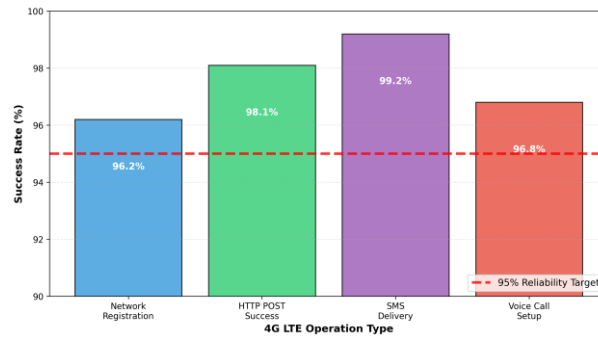


FIGURE 6.3 – A7670E LTE Cat-1 Module Performance Metrics (n=200)

GPS Positioning Accuracy Across Environments

Figure 6.4 assesses the accuracy of GNSS positioning in open sky, urban and indoor conditions. The horizontal error median was 8.5 m (IQR: 6.211.8 m) in open sky, 15.2 m (IQR: 10.420.6 m) in a city and 45.0 m indoors, where the signal obstruction significantly influenced the accuracy. A last-known-location caching mechanism is used to overcome poor indoor performance. In case of a decrease in signal quality to specified thresholds, location information is marked with a staleness indicator in emergency notifications, allowing responders to use other localization measures if required.

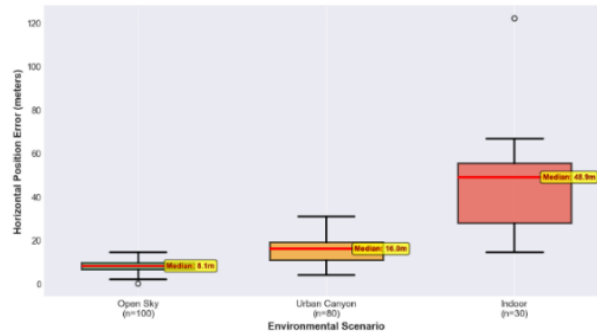


FIGURE 6.4 - GPS Positioning Accuracy Across Environmental Contexts

System Power Consumption Profile

Figure 6.5 presents the power distribution across major hardware components at a 5V supply. The Arduino® Nano™ 33 Bluetooth® Sense Rev2 draws 45 mA (225 mW), while the ESP32 with A7670E LTE Cat-1 GNSS module draws 380 mA (1900 mW), giving a peak system consumption of 2.125 W. Under continuous operation, a 3000 mAh Li-Po battery sustains approximately 7.1 hours of runtime. By enabling the ESP32 and LTE modules to deep sleep, using periodic edge detection, and duty cycling GNSS, the average power consumption drops to around 500 mW, allowing the device to run up to about 18 hours and ensuring it is suitable for long driving sessions.

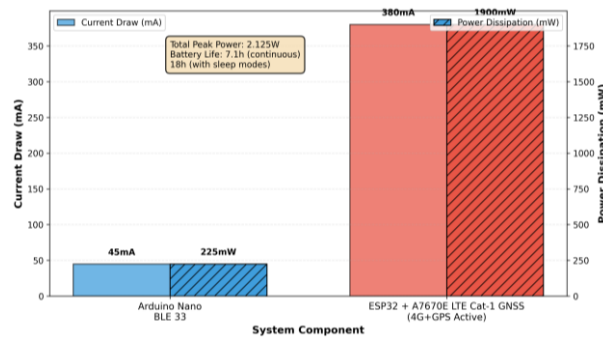


FIGURE 6.5 - System Power Consumption Profile at 5V Supply.

Feature Comparison Matrix

Figure 6.6 compares the proposed system with commercial wearables and research prototypes on six functional dimensions: on-device machine learning, cloud synchronisation, GNSS-based localisation, automated emergency notifications, mobile application support, and system-level integration. As indicated, the proposed framework is the one architecture that is considered to meet all six dimensions in one deployable unit. Wearables available in the market need smartphone connectivity and cloud-based analytics, which restrict autonomous emergency functionality. ML prototypes are research prototypes, which show capability but are not robustly integrated with clouds and automatic alerting. The suggested system provides a complete edge-IoT system with embedded cardiac intelligence, dual-mode emergency communications, real-time GNSS positioning, and cloud-synchronised remote monitoring in a vehicle-ready form factor.



FIGURE 6.6 - Multi-Dimensional Feature Comparison Matrix

CONCLUSION

The paper introduces a vehicle-integrated health monitoring system that integrates on-device ML inference, LTE emergency alerting, GNSS localisation, and cloud-synchronous mobile visualisation of real-time cardiac risk assessment. A7670E LTE Cat-1 module with ESP32 gateway allows reliable transmission of health data and automatic notifications, and a vibration motor provides a tactile driver feedback. The VitalFlow app is a remote monitoring application that is connected to the cloud. The system proves that the integration of embedded ML and IoT in cars can significantly improve road safety and emergency response outcomes.

REFERENCES

1. V. M G, R. Srikanth, and J. R, "Embedded machine learning for early detection of heart attack symptoms," in *Proceedings of the 9th International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud)*, (IEEE, 2025), pp. 570–583.
2. S. Nashif, Md.R. Raihan, Md.R. Islam, and M.H. Imam, "Heart Disease Detection by Using Machine Learning Algorithms and a Real-Time Cardiovascular Health Monitoring System," *WJET* **06**(04), 854–873 (2018).
3. S. Uma, and R. Eswari, "Accident prevention and safety assistance using IOT and machine learning," *J Reliable Intell Environ* **8**(2), 79–103 (2022).
4. R. Bharti, A. Khamparia, M. Shabaz, G. Dhiman, S. Pande, and P. Singh, "Prediction of Heart Disease Using a Combination of Machine Learning and Deep Learning," *Computational Intelligence and Neuroscience* **2021**(1), 8387680 (2021).
5. D. Zhang, Y. Chen, Y. Chen, S. Ye, W. Cai, J. Jiang, Y. Xu, G. Zheng, and M. Chen, "Heart Disease Prediction Based on the Embedded Feature Selection Method and Deep Neural Network," *Journal of Healthcare Engineering* **2021**, 1–9 (2021).
6. H. Elwahsh, E. El-shafeiy, S. Alanazi, and M.A. Tawfeek, "A new smart healthcare framework for real-time heart disease detection based on deep and machine learning," *PeerJ Computer Science* **7**, e646 (2021).
7. R. Alnashwan, M. Mashaabi, A. Alotaibi, H. Qudaih, and L. Albraheem, "IoT Based Accident Prevention System using Machine Learning techniques," in *2023 6th Artificial Intelligence and Cloud Computing Conference (AICCC)*, (ACM, Kyoto Japan, 2023), pp. 179–188.
8. L. Zhu, Y. Xiao, and X. Li, "Hybrid driver monitoring system based on Internet of Things and machine learning," in *2021 IEEE International Conference on Consumer Electronics and Computer Engineering (ICCECE)*, (IEEE, Guangzhou, China, 2021), pp. 635–638.
9. P. Sharma, and N. Sood, "Application of IoT and Machine Learning for Real-time Driver Monitoring and Assisting Device," in *2020 11th International Conference on Computing, Communication and Networking Technologies (ICCCNT)*, (IEEE, Kharagpur, India, 2020), pp. 1–7.
10. C.V.S. Babu, A. N. S., M.V. P., and R. Janapriyan, "IoT-Based Smart Accident Detection and Alert System:," in *Advances in Computational Intelligence and Robotics*, edited by P. Swarnalatha and S. Prabu, (IGI Global, 2023), pp. 322–337.
11. J.P. Li, A.U. Haq, S.U. Din, J. Khan, A. Khan, and A. Saboor, "Heart Disease Identification Method Using Machine Learning Classification in E-Healthcare," *IEEE Access* **8**, 107562–107582 (2020).
12. A. Garavand, F. Ameri, F. Salehi, A.H. Talebi, Z. Karbasi, and A. Sabahi, "A Systematic Review of Health Management Mobile Applications in COVID-19 Pandemic: Features, Advantages, and Disadvantages," *BioMed Research International* **2024**, 1–11 (2024).
13. Arduino S.r.l., "Arduino® Nano™ ESP32," (2023). Arduino and Nano are trademarks or registered trademarks of Arduino S.r.l.
14. DFRobot, "Gravity: I2C to Dual UART Module (DFR0627)," (2020).
15. Waveshare, "A7670E Cat-1/GNSS HAT," (2020).